Execution Report #5

The register runs dry

B5 (engine substitution), B7 (escalation band), B8 (specialization boundary), B9 (context paging), B4 (deontic fragment), and the consistency-radius theorem — the entire blocking-before-arXiv register, discharged

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Express lane. Six items, six scripts, all green at seed 20260816: the reputation market unravels without engine attestation and flips to the planner's rule with it (§1); the escalation debit has a unique separating threshold and a two-walled tuning band that can be honestly empty (§2); the whitepaper's specialization threshold is *falsified* and replaced by the exact Erlang-C boundary, with the succession rule priced (§3); context paging degrades only linearly under a corrupted pin oracle — if and only if pins are re-derived on touch (§4); commitment-conflict detection is polynomial inside the Horn fragment and NP-complete one step outside (§5); and the sheaf detector's residual is now a *certified lower bound* on the injected lie, with localization and complexity theorems (§6). With these, the proof-debt register's blocking-before-arXiv list (A3, A4, B5, B7, B8, B9) is clear. Every number is tagged [verified] (externally recomputable) or [internal] (regenerates from the named script, seed 20260816).

1 B5 — The engine behind the reputation

The story. An identity builds a glowing record running a strong engine, then quietly swaps a cheap one in behind the same name. Buyers see the reputation; they do not see the engine. This is the resume-button feature's named attack — and it is Akerlof's used-car market, run *inside one identity.*

The idea in one breath: unattested, the price can never depend on the actual engine, so swapping always pays and pooling on quality dies; attest the engine on every witnessed outcome and the substitution incentive flips to the planner's own rule — and the same three checks make resurrection across providers sanction-sound.

Theorem (unraveling). With engines $\theta_H > \theta_L$ at costs $c_H > c_L$ and the engine unobserved, the swap gain $(p - c_L) - (p - c_H) = \Delta c > 0$ is price-independent: no belief supports pooling on θ_H and nothing exists to separate on. Committed θ_H sellers participate iff $\mu \geq \mu^\star = (c_H - \theta_L)/\Delta\theta$; below it the death spiral rests at θ_L. Deterrence without attestation needs an audited stake $qW \geq \Delta c$.

Theorem (IC flip). With daemon-attested engine ids, reputation keys on (principal, engine) and the swap gain becomes $\Delta c - \Delta\theta$: substitution pays iff the weak engine is socially better, at zero audit stake.

Theorem (resurrection soundness). Sanction-respecting reputation (Def. III.6.1) survives provider migration under (i) credential+continuity-witness verification, (ii) engine attestation in successor outcomes, (iii) open commitments closed or renegotiated with escrow delta — and each dropped clause admits a shortest-trace escape: whitewash-by-resurrection, engine-history shedding, liability stranding.

Numbers. θ_H=1, θ_L=0.4, c_H=0.5, c_L=0.2: swap gain 0.3 at every price, $\mu^\star = 1/6$, attested keep 0.5 vs swap 0.2 [verified]. Sweeps: 4000 instances $\times$ 8 prices, 0 pooling survivals; thresholds match simulation exactly; 2000 heterogeneous spirals monotone to the Akerlof fixed point (μ_0=0.5 $\to$ 1/3; μ_0=0.2 $\to$ total unraveling); migration machine: 747 states to depth 7, **0** Def-III.6.1 violations intact; mutants caught in 1, 2, 2, 2 steps [internal, `b5_engine_substitution.py`].

The wrong turn, reported. A first pass keyed *sanctions* like scores, per (identity, engine) — and the intact protocol failed Def. III.6.1 in one step, no migration needed: the sibling engine key is a free escape hatch. The fix is the paper's one-liner: price per (principal, engine), sanction per principal — one rule closing engine substitution and skill-versioning Gap #4 together.

Honest boundary. Two periods, no continuation value beyond the stake; attestation soundness is a deployment (TEE) assumption, per the declined-claims list; the migration result is bounded small-model checking of the protocol spec, which the daemon must refine; Def. III.6.1 is score-only — escrow enters as attribution, not conservation.

2 B7 — The costly-escalation debit: threshold equilibrium and its tuning band

The story. An agent privately observes urgency $u \sim F$ on $[0, 1]$; escalating surfaces the item on the operator's reserved distress lane, where it is validated with probability $V(u)$ and yields benefit $b(u)$; a *dismissed* escalation debits standing by δ, worth w per unit of surfacing weight. Crying wolf has a price — but what price is right?

Monotone benefit and validation make the escalation payoff single-crossing, so a unique threshold type separates escalators from the silent; and between the alarm-fatigue wall and the distress-silencing wall the feasible debit is an interval — possibly empty, in which case no tuning saves an under-provisioned channel.

Theorem (threshold equilibrium; single crossing). With $\Pi_\delta(u) = b(u) - \delta w\,(1 - V(u))$, monotone b and V make Π_δ strictly increasing, so with $\Pi_\delta(0) < 0 < \Pi_\delta(1)$ there is a unique $u^\star(\delta)$ solving $b(u^\star) = \delta w\,(1 - V(u^\star))$, and the unique best response is "escalate iff $u \geq u^\star(\delta)$" — a separating equilibrium in the Spence mold. $u^\star$ is nondecreasing in δ with interior slope $w(1 - V(u^\star))/\big(b'(u^\star) + \delta w V'(u^\star)\big)$. For $u \sim \mathrm{Unif}[0, 1]$, $b(u) = u$, $V(u) = u$: $u^\star(\delta) = \delta w/(1 + \delta w)$.

Theorem (the tuning band). With alarm load $L(\delta) = \int_{u^\star(\delta)}^1 [c_{\mathrm{att}} + C_{\mathrm{fa}}(1 - V(u))]\, dF$ nonincreasing and miss loss $\mathrm{ML}(\delta) = C_{\mathrm{miss}}\big(F(u^\star(\delta)) - F(u_{\mathrm{crit}})\big)^+$ nondecreasing, the feasible set $\{\delta : L \leq A,\ \mathrm{ML} \leq M\}$ is an interval $[\delta_{\min}, \delta_{\max}]$, nonempty iff $L(\delta_{\max}) \leq A$; here $u_{\mathrm{crit}} = (c_{\mathrm{att}} + C_{\mathrm{fa}})/(C_{\mathrm{miss}} + C_{\mathrm{fa}})$ is exactly R3's inspect threshold.

Numbers. Closed forms match bisection ($u^\star = \delta w/(1+\delta w)$; Lambert-$W$ for the exponential family) [verified]; single crossing and monotone $u^\star(\delta)$: 0/4000 violations over random monotone families; the non-monotone attack breaks the threshold in 1172/2000 draws (witness: u = 0.681 escalates while u = 0.751 is silent) [internal, `b7_escalation_band.py`]. R3's constants (C_{miss}=100, c_{att}=1, C_{fa}=5), w=1, M=1: budget A = 3.3 gives $\delta \in [0.035, 0.072]$; budget A = 2.0 gives $\delta_{\min} = 0.396 > \delta_{\max} = 0.072$ — *empty.* The zero-debit mutant blows the fatigue budget ($L(0) = 3.5$, the Cvach habituation regime); $\delta = 1$ silences 0.44 of genuine-distress mass (miss loss 44.3) yet passes the fatigue wall alone — one-sided tuning would miss it [internal].

Honest boundary. Monotone V is the condition earning its keep: non-monotone validation produces multi-interval escalation sets — measure the validation curve before trusting any δ. Empty-band regimes are real; there the fix is attention capacity or evidence quality, not tuning. Static one-shot model: no reputation dynamics of w, no operator strategic dismissal, no flood externality (R7/A2's job).

3 B8 — The specialization boundary, and the price of the succession rule

The story. Should an invariant-critical function have one deeply skilled owner or a pool of generalists? The whitepaper proposed a threshold. The sweep ran first — and the proposed threshold is *false in both directions*: it certifies specialists who lose at low utilization and rejects specialists who win at high utilization.

The exact boundary is Erlang-C: sole ownership pays iff the skill premium clears $g(\rho, c) = c\rho + c(1-\rho)/(c(1-\rho) + C(c, \rho))$ — which saturates at c, never diverges; and once the owner's death-rate–times–succession-time product exceeds $D^\star$, no skill premium rescues sole ownership at all.

Theorem (queueing specialization boundary). Requests arrive Poisson(λ); a sole specialist serves as $M/M/1$ at rate μ_s, the pool as $M/M/c$ at μ_g each, utilization $\rho = \lambda/(c\mu_g)$; accountability value A, waiting cost w. With $C(c, \rho)$ the Erlang-C delay probability, sole ownership weakly dominates iff

$$\frac{\mu_s}{\mu_g} \;\geq\; g_A(\rho, c) \;=\; c\rho + \frac{1}{1 + \frac{C(c,\rho)}{c(1-\rho)} + \frac{A\mu_g}{w\lambda}},$$

reducing at $A = 0$ to $g(\rho, c) = c\rho + \frac{c(1-\rho)}{c(1-\rho)+C(c,\rho)}$, with $g(\rho, 1) = 1$, $g \to c$ as $\rho \to 1$, and $g(\rho, 2) = 1 + 2\rho - \rho^2$ in closed form.

Theorem (price of the succession rule). Specialist dies at rate ξ, succeeded after Exp(η) spin-up, work resuming, requests queueing throughout. With $\mu_{\text{eff}} = \mu_s\eta/(\xi + \eta)$: $W = \big(1 + \mu_s\xi/(\xi + \eta)^2\big)/(\mu_{\text{eff}} - \lambda)$ exactly, with infimum $W_\infty = \xi/(\eta(\xi + \eta))$ as $\mu_s \to \infty$: no skill buys back residual downtime. Pooling dominates at *every* skill premium iff $\xi/\eta > D^\star = \eta K/(1 - \eta K)$, $K = A/(w\lambda) + W_{\text{pool}}$.

Numbers. The proposed threshold $\tilde{g} = 1 + (c-1)\rho/(1-\rho)$ is falsified both directions: c=2, ρ=0.1 false-certify (sim $z = -15.7$); c=2, ρ=0.5 false-reject ($z = +32$); crossing at $\rho = (3 - \sqrt{5})/2 \approx 0.382$; $\tilde{g}$ diverges where g saturates (ρ=0.9, c=8: $g = 7.73$ vs $\tilde{g} = 64$) [internal, `b8_specialization.py`]. W closed form vs matrix-geometric $2{\times}10^{-13}$, vs truncated CTMC 10^{-14}, vs Gillespie $|z| \leq 0.8$; 60-instance sweep, 0 sign violations; succession instance $\eta = 0.25$: $D^\star = 0.350$, and at $\xi/\eta = 2$ even $\mu_s = 10^8$ loses [internal]. The breakdown-forgetting mutant is caught by the high-ξ canary. A common misread to preempt: $D^\star$ does not say specialists are bad — it prices exactly when a succession plan (large η) keeps sole ownership viable.

The wrong turn, reported. The naive decomposition $W = 1/(\mu_{\text{eff}} - \lambda) + W_\infty$ is refuted (4.17 vs true 8.17 at the canary instance): downtime also builds queue, amplifying the residual by $\mu_{\text{eff}}/(\mu_{\text{eff}} - \lambda)$.

Honest boundary. Mean-cost comparison only (no tail/SLA percentiles); FCFS, exponential everything; breakdowns-at-all-times with preemptive resume (the death/heartbeat model); A and w are policy prices the theorem constrains, not picks.

4 B9 — Context paging with an untrusted pin oracle

The story. The resident context is a cache of spans; a feature oracle marks some spans load-bearing (pinned). The oracle is sometimes wrong — at rate φ, the same forgeable-feature budget R1 already carries. How badly does paging degrade?

Linearly, and only linearly: with Young's Landlord on the honestly-unpinned region and pins re-derived on every fetch, a corrupted consultation costs at most one refetch — but trust a pin blindly and a single corrupted bit costs $\Theta(N)$.

Theorem (import; Young 2002). Landlord with cache k over spans of size $s(f)$ and refetch cost $c(f)$ satisfies $\mathrm{cost}_{LL}(\sigma) \le \frac{k}{k-h+1}\,\mathrm{OPT}_h(\sigma)$ for every $h \le k$ — the weighted file-caching transfer variable-size context spans need.

Theorem (linear φ-degradation). Unit cost density $c(f) = s(f)$ (the context-window regime); honest pinned set P; oracle corrupted on C consultations ($\mathbb{E}[C] = \varphi N$, obliviously), pins re-derived on every fetch (*repair-on-touch*). Then pin-augmented Landlord pays, for every $p \le h \le k$,

$$\mathrm{Cost} \;\le\; c(P) + \frac{k-p}{k-h+1}\,\mathrm{OPT}_{h-p}(\sigma_{\mathrm{unpinned}}) + \varphi N c_{\mathrm{refetch}},$$

an *additive*, linear φ term: each corruption perturbs one eviction pass and is charged at most one refetch (missed pin: repair-on-touch buys it back; mis-pinned junk: the displaced volume refetches for at most $s(g) \le c_{\mathrm{refetch}}$).

Numbers. Import vs exact OPT (subset-DP): 122 (instance, h) pairs, 0 violations, cyclic thrash instances at exact equality [internal, `b9_context_paging.py`]. Degradation: 50 instances at $\varphi \in \{0.1, \ldots, 0.4\}$, all h, 0 violations; cost-vs-φ slope 22.8 vs ceiling $N c_{\mathrm{refetch}} = 1200$ (1.9%; $R^2 = 0.994$); per-corruption charge $\le 0.100\, c_{\mathrm{refetch}}$ over 120 trials. An *adaptive* corruptor that sees cache state maxed at $0.778\, c_{\mathrm{refetch}}$ per corruption — inside the bound; linearity did not break. Mutation: blind pin-trust turns *one* corruption into 317 refetch-equivalents at N=1920 vs the graceful policy's exactly one refetch [internal].

The wrong turn, reported. A corruption can *help* (extra cost < 0) when the falsely-unpinned span was crowding a thrashing working set — the bound is one-sided by design; the final mutation scenario isolates the pin effect.

Honest boundary. Unit cost density is load-bearing for the mis-pin charge. Repair-on-touch is *the* load-bearing guard — without it damage is unbounded (staleness is priced charitably at refetch cost; acting on absent load-bearing context is a correctness failure). The oblivious hypothesis was not the boundary for the additive term (greedy-adaptive stayed inside; the oblivious/adaptive gap lives in the multiplicative $O(\log k)$ of randomized marking); a stronger adaptive adversary is untested. The sweep is evidence at small scale; the second-order credit step rests on the Landlord term's slack, verified per instance.

5 B4 — The tractable conflict fragment and its frontier

The story. The condominium harbor's lookout must answer, at proposal time, whether an incoming commitment conflicts with the accepted set. Cheap detection is a language-design choice: the commitment language buys it by construction.

Inside Horn rules, ground intervals, and difference constraints, conflict detection is a polynomial, witness-producing decision; allow one disjunctive obligation and it is NP-complete — the fragment boundary is not taste but the exact price of proposal-time checking.

Theorem (membership). In the fragment $\mathcal{L}_c$ (facts; definite Horn rules with $\bot$ heads allowed; deontic rules body $\to \mathsf{M}(\text{act}, s, [t_1, t_2])$, $\mathsf{M} \in \{\mathsf{O}, \mathsf{F}\}$; exclusive interval claims; difference constraints $x_j - x_i \leq c$), conflict detection is decidable in $O(H + T \log T + C \log C + V \cdot E)$ with a polynomial witness per conflict: Dowling–Gallier unit propagation computes the least Horn model (= statuses forced in every model, by monotonicity); a keyed sweep-line finds every O/F and claim overlap; Bellman–Ford returns a negative cycle iff the deadline system is infeasible.

Theorem (frontier). Extend $\mathcal{L}_c$ with disjunctive obligations $\mathsf{O}(l_1 \vee \cdots \vee l_k)$ under discharge-choice semantics: conflict-freedom is NP-complete (membership via the selection certificate; hardness from 3-SAT — clause $\mapsto$ disjunctive obligation over selector literals, variable $\mapsto$ an O/F pair on identical scope and interval).

Numbers. 3000 random in-fragment policy sets vs a brute-force oracle (all 2^5 Horn models intersected; exhaustive integer search over the complete potential box): 885 conflicts (121 $\bot$, 195 O/F, 484 claims, 213 temporal), 149 reachable only through Horn propagation, **0 disagreements**; least model = intersection-of-all-models on all 3000 [internal, `b4_deontic_fragment.py`]. The reduction verified both directions on 8 SAT + 8 UNSAT instances, 16/16. Mutation: removing Horn propagation misses the four-line indirect witness (deploy $\to$ prod access $\to$ obliged write vs standing prohibition) and 100/885 sweep conflicts — propagation is load-bearing at scale.

Honest boundary. Completeness is relative to least-model semantics (monotone bodies; negation-as-failure leaves the fragment). Ground atoms and intervals only — unification and symbolic endpoints coupled to difference variables are outside what is proven. The batch bound is proven; the incremental form is a constants improvement. The checker certifies the norm *system*, not the norms: a bad policy consistently held is consistent.

6 The consistency-radius theorem (W8 tail)

The story. The R6 harness validated the completion residual r statistically. What was missing was the theorem: what exactly does $r > 0$ prove, how large a lie does it certify, where does it point, and what does it cost to compute?

r is the exact minimum edge-space perturbation any adversary must have injected, it localizes to cycles through the equivocator, and it computes as at most L scalar graph-Laplacian solves — with the single-equivocator condition now proved to be the honest scope, because coalitions on a cycle can cancel to zero.

Theorem CR-1 (soundness). $r > 0$ iff no global section explains the compared+relayed data; honest executions give $r = 0$ exactly, for every visibility pattern. Every offset pattern consistent with the data satisfies $\|\varepsilon_K\| \geq r$, with equality at the completion. Single equivocator q: $\sigma^+_{\min}(M_q)\,\mathrm{dist}(o, \ker M_q) \leq r \leq \mathrm{dist}(o, \ker M_q)$, and for a single-edge lie of size s in coordinate c, $r = |s|\sqrt{1 - R^{G_c}_{\text{eff}}(e)}$ — on C_n, $|s|/\sqrt{n}$: the harness's 1.2247 *is* $3\sqrt{1 - 5/6}$.

Theorem CR-2 (localization). Noise-free, single equivocator: the residual's support lies on cycles of the coordinate subgraphs through q (electrical-flow support argument) — the harness's 200/200 is now a theorem; with noise, guaranteed while $2\|\eta_K\| < \max_f \|(\Pi_K \varepsilon_K)_f\|$.

Theorem CR-3 (complexity). r is one sparse least-squares that decomposes into at most L scalar graph-Laplacian systems: $\widetilde{O}(|E| L \log(1/\epsilon))$ with SDD solvers. (Bach's #P-hardness concerns coherent sheaves on projective space, not finite cellular sheaves over

ℝ.)

Numbers. Soundness 0/600 violations, achievability 600/600; the effective-resistance closed form exact on C_4–C_{12} and 200 random graphs; the $\sigma^+_{\min}$ lower bound tight under the adversarial offset (0.4082); a uniform (kernel) lie of norm 5 gives $r = 2{\times}10^{-14}$ — invisible in principle, correctly not flagged [internal, `sheaf_consistency_radius.py`]. Localization 200/200 (two-cycles topology) + 128/128 (severed expanders); the noise boundary is measured: $\geq 95\%$ up to $\sigma_\eta/\sigma_{\text{eq}} \approx 0.1$, 76.5% at 0.2. Timing slope vs $|E|$: CG 0.65 (linear-ish), dense 1.88. The sweep found the promised crack: two coordinated liars on a common cycle cancel to $r = 6{\times}10^{-15}$ while each alone is detectable at $|s|/\sqrt{8}$ — the detection bound is single-equivocator; soundness (r never overstates) is unconditional. The D2-reintroduction mutant fires on a severed lie exactly as CR-1(i) predicts.

Honest boundary. Detection and localization are single-equivocator: coalition-coboundary lies are consistent counterfactual worlds, provably dark. r attributes nothing (a $+o$ lie by q toward w and a $-o$ lie by w toward q yield identical observations — signatures attribute, cohomology localizes). Vanishing r is not an all-clear (uniform lies, severed edges, the Carù abelianization gap). Under noise, soundness needs a threshold above the honest floor.

7 Program status

Item	Status	Headline	Feeds
A1–A4, A6, A7, B1–B3, B6, C0, C1	done	Execution Reports 1–4	Papers 1–4, 8
Zoom theorem; sheaf harness	done	Paper 1 closed; verdict COMMIT	Papers 1, 7
B5	**done**	unraveling; IC flip; resurrection soundness	Paper 5
B7	**done**	threshold equilibrium; tuning band	Paper 1 orbit
B8	**done**	exact Erlang-C boundary; $D^\star$ succession price	Paper 6
B9	**done**	Landlord import; linear φ-degradation	Paper 1 orbit
B4	**done**	polynomial fragment; NP-complete frontier	Paper 6
CR-1/2/3	**done**	r certifies, localizes, computes	Paper 7

The blocking-before-arXiv register (A3, A4, B5, B7, B8, B9) is **clear**. Remaining: Papers 5–7 assemblies (all inputs now in hand), C2 (minimum-viable take-rate), the lifts (C0 $\to$ Apalache, C1 $\to$ Isabelle), and the whitepaper folds (thm:specialization must adopt the exact g; the context-paging and escalation status notes upgrade to executed).

Now you try. Compute $D^\star$ for a sole-owned role in your own deployment: measure its request rate λ, the pool's service rate μ_g, the role's historical death rate ξ (departures, deprecations), and your successor spin-up time $1/\eta$; if ξ/η exceeds $\eta K/(1-\eta K)$, no skill premium rescues sole ownership — write the succession plan first.

Reproducibility. Six self-contained scripts (`b5_engine_substitution.py`, `b7_escalation_band.py`, `b8_specialization.py`, `b9_context_paging.py`, `b4_deontic_fragment.py`, `sheaf_consistency_radius.py`); every [internal] number regenerates deterministically at seed 20260816, every mutation prints its witness, and each script exits nonzero on any failed check. All six run in the harbor-results-estate CI job.